BADR UNIVERSITY IN CAIRO (BUC) — FACULTY OF ENGINEERING

Fluid Power Engineering (Dr. M. Galal Rabie) — Theoretical Background & Master Textbook Solutions

Course: MEC 431 / MEC 451 / MEC 457 | **Examiner:** Dr. Ziad Ahmed Ibrahim | **Teaching Assistant:** Eng. Moh Zedan
Complete Mathematical Derivations & Advanced Textbook Solved Problems

Part 1: Lumped Parameter Modeling of Hydraulic Transmission Lines (R, I, C)

1.1 Hydraulic Resistance (R_{hyd}) and Equivalent Networks

In laminar flow, fluid friction creates a pressure drop directly proportional to flow rate:

$$\Delta P = \frac{128\mu L}{\pi D^4} \cdot Q = R_{hyd} \cdot Q \qquad \text{where } R_{hyd} = \frac{128\mu L}{\pi D^4} \quad \left[\frac{\text{Pa}\cdot\text{s}}{\text{m}^3}\right]$$

- **Lines Connected in Series:** Common flow rate Q. Total pressure drop is the sum of drops:

$$P_1 - P_3 = (P_1 - P_2) + (P_2 - P_3) = R_1 Q + R_2 Q = (R_1 + R_2)Q \implies \boxed{R_{series} = R_1 + R_2}$$

- **Lines Connected in Parallel:** Common pressure drop ΔP. Total flow is the sum of branch flows:

$$Q = Q_1 + Q_2 = \frac{\Delta P}{R_1} + \frac{\Delta P}{R_2} = \Delta P\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \implies \boxed{\frac{1}{R_{parallel}} = \frac{1}{R_1} + \frac{1}{R_2} \iff R_{parallel} = \frac{R_1 R_2}{R_1 + R_2}}$$

1.2 Hydraulic Inertia (I_{hyd}) and Inertial Networks

When fluid in a line accelerates, its mass resists velocity changes. Applying Newton's Second Law ($F = ma$) to an oil column of density ρ, area A, and length L:

$$\Delta P \cdot A = (\rho A L) \cdot \frac{dv}{dt} \implies \Delta P \cdot A = (\rho A L) \cdot \frac{1}{A}\frac{dQ}{dt} \implies \Delta P = \left(\frac{\rho L}{A}\right)\frac{dQ}{dt}$$

$$\boxed{\Delta P = I_{hyd} \cdot \frac{dQ}{dt}} \qquad \text{where} \qquad \boxed{I_{hyd} = \frac{\rho L}{A} = \frac{4\rho L}{\pi D^2}} \quad \left[\frac{\text{kg}}{\text{m}^4}\right]$$

- **Series Lines:** $\Delta P = \Delta P_1 + \Delta P_2 = I_1 \frac{dQ}{dt} + I_2 \frac{dQ}{dt} \implies \boxed{I_{series} = I_1 + I_2}$
- **Parallel Lines:** $\frac{dQ}{dt} = \frac{dQ_1}{dt} + \frac{dQ_2}{dt} = \frac{\Delta P}{I_1} + \frac{\Delta P}{I_2} \implies \boxed{\frac{1}{I_{parallel}} = \frac{1}{I_1} + \frac{1}{I_2} \iff I_{parallel} = \frac{I_1 I_2}{I_1 + I_2}}$

1.3 Hydraulic Capacitance (C_{hyd}) and Wall Elasticity

Fluid compressibility allows a volume of oil to store fluid when pressure rises. By definition:

$$Q_C = \frac{dV}{dt} = \frac{V_0}{B}\frac{dP}{dt} = C_{hyd} \cdot \frac{dP}{dt} \qquad \text{where for rigid pipes, } \boxed{C_{hyd} = \frac{V_0}{B} = \frac{\pi D^2 L}{4B}}$$

When pipe wall elasticity is taken into account under internal pressure ΔP:

Condition	Combined Hydraulic Capacitance Formula
1. Rigid Pipe Wall (Oil only)	$C = \frac{V}{B} = \frac{\pi D^2 L}{4B}$
2. Radial Wall Deformation Considered	$C = \frac{\pi D^2 L}{4}\left(\frac{1}{B} + \frac{D}{Eh}\right) = V\left(\frac{1}{B} + \frac{D}{Eh}\right)$
3. Both Radial & Axial Deformation Considered	$C = \frac{\pi D^2 L}{4}\left(\frac{1}{B} + \frac{5D}{4Eh}\right) = V\left(\frac{1}{B} + \frac{5D}{4Eh}\right)$

- **Capacitors in Parallel:** Common pressure $\frac{dP}{dt}$ → $\boxed{C_{parallel} = C_1 + C_2}$
- **Capacitors in Series:** Common flow Q → $\boxed{\frac{1}{C_{series}} = \frac{1}{C_1} + \frac{1}{C_2} \iff C_{series} = \frac{C_1 C_2}{C_1 + C_2}}$

Part 2: Equivalent Bulk Modulus of an Oil-Air Mixture (B_e)

Physical Problem: Hydraulic oil often contains entrained microscopic air bubbles. Since air is thousands of times more compressible than oil, even tiny volumetric air fractions ($\alpha \approx 1\%$) severely reduce the effective stiffness of the fluid, making actuators sluggish and causing servo instability.

Mathematical Derivation:

Let total mixture volume be V_T, where air volume is $V_g = \alpha V_T$ and pure oil volume is $V_o = (1-\alpha)V_T$.
The compression of the gas follows the polytropic gas law:

$$P \cdot V_g^n = \text{constant} \implies V_g^n dP + nV_g^{n-1} P dV_g = 0 \implies \frac{dP}{dV_g/V_g} = -nP \implies B_g = nP$$

Under a pressure increment ΔP, the volume changes of gas and oil are:

$$\Delta V_g = -\frac{V_g \Delta P}{B_g} = -\frac{\alpha V_T \Delta P}{nP}$$

$$\Delta V_o = -\frac{V_o \Delta P}{B_o} = -\frac{(1-\alpha) V_T \Delta P}{B_o}$$

The equivalent bulk modulus B_e is defined as:

$$B_e = -\frac{\Delta P}{\frac{\Delta V_T}{V_T}} = -\frac{\Delta P}{\frac{\Delta V_g + \Delta V_o}{V_T}} = \frac{\Delta P}{\frac{\alpha \Delta P}{nP} + \frac{(1-\alpha)\Delta P}{B_o}}$$

Equivalent Bulk Modulus Formula:

$$\boxed{\frac{1}{B_e} = \frac{\alpha}{nP} + \frac{1-\alpha}{B_o}} \quad \Longleftrightarrow \quad \boxed{B_e = \frac{nPB_o}{nP(1-\alpha) + \alpha B_o}}$$

(Notice: At high pressure $P \gg$, the term $\frac{\alpha}{nP} \to 0$ and $B_e \to B_o$. At low pressure, B_e drops catastrophically!).

Part 3: Valve Dynamics & Spool Transfer Function (Derivation 14)

3.1 Equation of Motion

Consider a hydraulic directional/servo spool of mass m, supported by centering springs of total stiffness k, moving in a sleeve with radial clearance c and wetted land area πDL, actuated by a net differential pressure force $f = \Delta P \cdot A_p$.

Applying Newton's Second Law:

$$m\frac{d^2x}{dt^2} + f_v\frac{dx}{dt} + kx = \Delta P \cdot A_p$$

where the viscous friction damping coefficient is $f_v = \frac{\pi\mu DL}{c}$.

3.2 Laplace Transformation & Transfer Function

Taking the Laplace transform assuming zero initial conditions:

$$(ms^2 + f_v s + k)X(s) = A_p \cdot P(s)$$

$$\boxed{G(s) = \frac{X(s)}{P(s)} = \frac{A_p}{ms^2 + f_v s + k} = \frac{K}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n}s + 1}}$$

Standard Second-Order Dynamic Parameters:

Static Gain: $\boxed{K = \frac{A_p}{k}}$ $[\text{m/Pa}]$

Undamped Natural Frequency: $\boxed{\omega_n = \sqrt{\frac{k}{m}}}$ $[\text{rad/s}]$

Damping Ratio: $\boxed{\zeta = \frac{f_v}{2\sqrt{km}} = \frac{\pi\mu DL}{2c\sqrt{km}}}$

3.3 Numerical Example from Textbook (Derivation 14b):

Given: $m = 0.07$ kg, $D = 8$ mm, $L = 40$ mm, $c = 4$ μm, $k = 20$ kN/m, $\mu = 0.02$ Pa · s, $\Delta P = 4$ bar $= 4 \times 10^5$ Pa.

$$A_p = \frac{\pi}{4}(0.008)^2 = 5.027 \times 10^{-5}\ \text{m}^2$$

$$f_v = \frac{\pi \times 0.02 \times 0.008 \times 0.040}{4 \times 10^{-6}} = 5.027\ \text{N} \cdot \text{s/m}$$

$$\omega_n = \sqrt{\frac{20,000}{0.07}} = 534.5\ \text{rad/s} \quad (85.1\ \text{Hz})$$

$$\zeta = \frac{5.027}{2\sqrt{20,000 \times 0.07}} = \frac{5.027}{2 \times 37.416} = 0.0672 \quad \text{(Highly underdamped! Oscillatory response!)}$$

Steady-state spool displacement:

$$x_{ss} = K \cdot \Delta P = \frac{A_p \Delta P}{k} = \frac{5.027 \times 10^{-5} \times 4 \times 10^5}{20,000} = 0.001005\ \text{m} = \mathbf{1.01\ mm}$$

Part 4: Rotary Spool Viscous Friction (Derivation 12b)

When a cylindrical spool of radius r and length L rotates at angular speed ω inside a sleeve of radius r_1 with clearance $c = r_1 - r$:

Surface linear velocity: $u = r\omega$. Velocity gradient in thin gap: $\frac{du}{dr} = \frac{r\omega}{c}$.
Shear stress: $\tau = \mu\frac{du}{dr} = \frac{\mu r\omega}{c}$.
Resisting viscous friction torque:

$$T = \tau \cdot A_s \cdot r = \left(\frac{\mu r\omega}{c}\right) \cdot (2\pi rL) \cdot r = \left(\frac{2\pi\mu r^3 L}{c}\right)\omega = f_\omega \cdot \omega$$

Rotary Viscous Damping Coefficient:

$$\boxed{T = f_\omega \cdot \omega} \qquad \text{where} \qquad \boxed{f_\omega = \frac{2\pi\mu r^3 L}{c}} \quad [\mathrm{N \cdot m \cdot s/rad}]$$

Part 5: Advanced Solved Textbook Pump Problems (Problems 25, 26, 27, 28)

Problem 25: External Gear Pump Comprehensive Sizing

Given Data: $n = 1450$ rpm $= 24.167$ rps, $z = 12$, $m = 3.5$ mm $= 0.0035$ m, $b = 20$ mm $= 0.02$ m, $\alpha = 20^\circ$, $P_i = 0.2$ MPa, $P_o = 15$ MPa, $\eta_m = 0.85$, $\eta_v = 0.90$, $\eta_h = 1.0$.

1. Geometric Volume (Dr. Rabie Formula):

$$V_g = 2\pi b m^2 (z + \sin^2\alpha) = 2\pi(0.02)(0.0035)^2(12 + \sin^2 20^\circ)$$

$$\sin 20^\circ = 0.34202 \implies \sin^2 20^\circ = 0.1170$$

$$V_g = 2\pi(0.02)(1.225 \times 10^{-5})(12.117) = \mathbf{1.865 \times 10^{-5}\ m^3 = 18.65\ cm^3/rev}$$

2. Delivery Flow Rates:

$$Q_t = V_g \cdot n = (1.865 \times 10^{-5}) \times 24.167 = 4.507 \times 10^{-4}\ \text{m}^3/\text{s} = \mathbf{27.04\ L/min}$$

$$Q = Q_t \cdot \eta_v = (27.04) \times 0.90 = \mathbf{24.34\ L/min} = 4.056 \times 10^{-4}\ \text{m}^3/\text{s}$$

3. Delivery Pulsation Coefficient (σ):

$$\sigma = \frac{\pi^2 \cos^2\alpha}{4(z+1)} = \frac{\pi^2 \cos^2 20^\circ}{4(12+1)} = \frac{8.7148}{52} = 0.1676 = \mathbf{16.8\%}$$

4. Leakage Flow Rate & Leakage Resistance:

$$Q_L = Q_t - Q = 4.507 \times 10^{-4} - 4.056 \times 10^{-4} = 4.51 \times 10^{-5}\ \text{m}^3/\text{s}$$

$$\Delta P = 15 \times 10^6 - 0.2 \times 10^6 = 14.8 \times 10^6\ \text{Pa}$$

$$R_L = \frac{\Delta P}{Q_L} = \frac{14.8 \times 10^6}{4.51 \times 10^{-5}} = \mathbf{3.28 \times 10^{11}\ Pa \cdot s/m^3 = 328\ GNs/m^5}$$

5. Driving Torque & Input Power:

$$T = \frac{V_g(P_o - P_i)}{2\pi\eta_m\eta_h} = \frac{(1.865 \times 10^{-5}) \times (14.8 \times 10^6)}{2\pi \times 0.85 \times 1.0} = \frac{276.02}{5.3407} = \mathbf{51.69\ N \cdot m}$$

$$N_{in} = 2\pi n T = 2\pi \times 24.167 \times 51.69 = \mathbf{7.848\ kW}$$

$$N_h = P_o \cdot Q = (15 \times 10^6) \times (4.056 \times 10^{-4}) = \mathbf{6.085\ kW}$$

6. Volumetric Efficiency at Increased Pressure ($P = 220$ bar $= 22$ MPa):

$$\Delta P_{220} = 22 \times 10^6 - 0.2 \times 10^6 = 21.8 \times 10^6\ \text{Pa} \approx 22 \times 10^6\ \text{Pa}$$

$$\eta_{v,220} = 1 - \frac{\Delta P_{220}}{R_L \cdot V_g \cdot n} = 1 - \frac{22 \times 10^6}{(3.28 \times 10^{11}) \times (1.865 \times 10^{-5}) \times 24.167} = 1 - 0.1488 = \mathbf{0.851 = 85.1\%}$$

Problem 26: Maximum Allowable Speed of Gear Pump (Cavitation Limit)

Problem Statement: Calculate the maximum allowable rotational speed of the gear pump from Problem 25 connected to a suction line: $L = 1$ m, $d = 13$ mm, oil head $H = 0.3$ m, tank pressure $P_T = 0.13$ MPa, $\rho = 870$ kg/m^3,

$\lambda = 0.035$.

Tip speed cavitation limit relates inlet pressure P_i to circumferential tooth speed $u = 2\pi n r$:

$$u = \sqrt{\frac{2P_i}{\rho}} \implies 2\pi n r = \sqrt{\frac{2P_i}{\rho}} \implies n = \frac{1}{2\pi r}\sqrt{\frac{2P_i}{\rho}} = \frac{1}{\pi r}\sqrt{\frac{P_i}{2\rho}}$$

Tip radius: $r = \frac{mz}{2} + m = 0.5(0.0035)(12) + 0.0035 = \mathbf{0.0245\ m}$.

Through iterative convergence between pipe loss $\Delta P = \lambda \frac{L}{d}\frac{\rho v^2}{2}$ and pump suction pressure $P_i = P_T + \rho g H - \Delta P$:

$$\mathbf{n = 63\ rps = 3778\ rpm} \quad (v = 8.85\ \text{m/s},\ \Delta P = 91.7\ \text{kPa},\ P_i = 40.87\ \text{kPa})$$

Problem 27: Bent Axis Axial Piston Pump

$z = 9, d = 9.3\ \text{mm}, D = 33\ \text{mm}, n = 4000\ \text{rpm}, P_i = 0.3\ \text{MPa}, P = 18\ \text{MPa}, \eta_v = 0.94, \eta_t = 0.89, \gamma = 20^\circ$:

$$V_g = \frac{\pi}{4} d^2 D z \sin\gamma = \frac{\pi}{4}(0.0093)^2(0.033)(9)\sin 20^\circ = \mathbf{6.90 \times 10^{-6}\ m^3 = 6.90\ cm^3}$$

$$Q_t = V_g \cdot n = (6.90 \times 10^{-6}) \times (4000/60) = 4.60 \times 10^{-4}\ \text{m}^3/\text{s} = \mathbf{27.60\ L/min}$$

$$Q = Q_t \cdot \eta_v = 4.32 \times 10^{-4}\ \text{m}^3/\text{s} = \mathbf{25.94\ L/min}$$

$$\eta_m = \frac{\eta_t}{\eta_v} = \frac{0.89}{0.94} = 0.9468 \implies T = \frac{V_g(P - P_i)}{2\pi\eta_m} = \frac{(6.90 \times 10^{-6})(17.7 \times 10^6)}{2\pi \times 0.9468} = \mathbf{20.53\ N \cdot m}$$

$$N_m = \frac{Q(P - P_i)}{\eta_t} = \frac{(4.324 \times 10^{-4})(17.7 \times 10^6)}{0.89} = \mathbf{8599\ W = 8.60\ kW}$$

Leakage: $Q_L = 2.76 \times 10^{-5}\ \text{m}^3/\text{s} \implies R_L = \frac{17.7 \times 10^6}{2.76 \times 10^{-5}} = \mathbf{6.412 \times 10^{11}\ Pa \cdot s/m^3}$.

At $P = 30\ \text{MPa}$: $Q_{L,30} = \frac{30 \times 10^6}{6.412 \times 10^{11}} = 4.68 \times 10^{-5}\ \text{m}^3/\text{s} \implies Q_{30} = \mathbf{24.78\ L/min}$, $T_{30} = \mathbf{34.80\ N \cdot m}$.

Problem 28: Swash Plate Axial Piston Pump

$z = 7, d = 10\ \text{mm}, D = 35\ \text{mm}, \gamma = 20^\circ, n = 3000\ \text{rpm}, \eta_m = 0.9, \eta_h = 0.99, P_i = 0, R_L = 258\ \text{GNs/m}^5$:

$$V_g = \frac{\pi}{4} d^2 D z \tan\gamma = \frac{\pi}{4}(0.01)^2(0.035)(7)\tan 20^\circ = \mathbf{7.00 \times 10^{-6}\ m^3 = 7.00\ cm^3}$$

$$Q_t = (7.00 \times 10^{-6}) \times (3000/60) = 3.50 \times 10^{-4}\ \text{m}^3/\text{s} = \mathbf{21.00\ L/min}$$

At $P = 300\ \text{bar} = 30\ \text{MPa}$:

$Q_{300} = Q_t - \frac{30 \times 10^6}{2.58 \times 10^{11}} = 3.50 \times 10^{-4} - 1.163 \times 10^{-4} = \mathbf{2.34 \times 10^{-4}\ m^3/s = 14.04\ L/min}$.

At $P = 10\ \text{MPa}$: $Q_{10} = 3.11 \times 10^{-4}\ \text{m}^3/\text{s} \implies \eta_{v,10} = 0.89 \implies \eta_{T,10} = 0.89 \times 0.9 \times 0.99 = \mathbf{0.791 = 79.1\%}$.

Maximum closed-line pressure: $P_L = Q_t \cdot R_L = (3.50 \times 10^{-4}) \times (2.58 \times 10^{11}) = \mathbf{90.3\ MPa = 903\ bar}$.